

UNIVERSIDAD EAFIT
School of Applied Sciences and Engineering
Department of Mechanical Engineering
IM0240 – Machine Design I

Drawing package

Tripteron CNC router machine

Delivery 2 – English cover and drawing index

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1. About this package

This package contains the complete set of manufacturing and assembly drawings of the Tripteron CNC router. The drawing sheets that follow this index are the CAD sheets as issued from SolidWorks; they are reproduced unchanged so that the dimensions, tolerances and revision state shown on them remain the ones the parts were made to.

General notes applying to every sheet:

- **All dimensions are in millimetres (mm).** Angles are in degrees.
- Projection: first-angle (ISO-A).
- Sheet format: A3 unless stated otherwise on the sheet.
- Unless otherwise indicated: linear tolerance ± 0.5 mm, angular tolerance $\pm 0.5^\circ$, break all sharp edges.
- Do not scale the drawing; work from the stated dimensions.
- Materials are stated in the title block of each sheet; the structural members are steel tube and plate, the links are $25 \times 25 \times 2$ mm square steel tube.

2. Drawing index

Sheet	Title	Scale	Group
Group 1 – Chassis and structure			
1	Structural table – welded frame	1:10	Chassis
2	Plate, 1/8 in – table top	1:5	Chassis
3	Top support 1	1:2	Chassis
4	Motor support	1:2	Chassis
5	Plate, 1/4 in – reinforcement	1:1	Chassis
6	Motor-side support	1:1	Chassis
7	Auxiliary top support	1:1	Chassis
8	Auxiliary support 2	1:1	Chassis
Group 2 – Electrical system			
9	Control wiring – microcontroller and stepper drivers	1:1	Electrical
10	Wiring diagram, sheet 2	1:1	Electrical
11	Wiring diagram, sheet 3	1:1	Electrical
12	Wiring diagram, sheet 4	1:1	Electrical
13	Wiring diagram, sheet 5	1:1	Electrical
Group 3 – Guide-supported axis system			
14	Guide system – general assembly	1:3	Guides
15	Guide system – assembly, second view	1:4	Guides
16	Bearings, motor and lead screw	1:1	Guides
17	Guide rail assembly, sheet 1	1:3	Guides

Sheet	Title	Scale	Group
18	Guide rail assembly, sheet 2	1:3	Guides
19	Guide rail assembly, sheet 3	1:3	Guides
20	Guide carriage – detail	2:1	Guides
21	Guide carriage – views	1:1	Guides
22	Bearing support – views	1:1	Guides
Group 4 – Arms (R-pair system)			
23	R-pair system – component list, qty 3	1:1	Arms
24	Bearing housing tube	2:1	Arms
25	Arm link with internal bore, qty 6	2:1	Arms
26	Arm component, qty 9	2:1	Arms
27	Arm component, qty 3	2:1	Arms
28	Arm component, qty 9	2:1	Arms
Group 5 – Tool support			
29	Tool (spindle) support	1:1	Tool
Group 6 – Assembled machine			
30	Complete machine – general assembly	1:10	Assembly
31	Complete machine – views	1:1	Assembly
32	Washer (spacer)	10:1	Assembly
33	Internal spacer	5:1	Assembly
34	Spacer – section A–A	2:1	Assembly
35	Spacer – views	2:1	Assembly

3. Relation to the other deliveries

Group	Where it is verified
Chassis and structure	Delivery 3, Sheet 4: finite element analysis of the welded frame (safety factor 66, maximum deflection 0.102 mm).
Guide-supported axes	Delivery 3, Sheet 3: lead-screw torque, efficiency, stresses and column buckling (safety factor 3.21 against buckling).
Arms	Delivery 3, Sheet 2: internal loads and bending stress in the six links (maximum 26.1 MPa, safety factor 9.6).
Assembled machine	Delivery 3, Sheet 1: kinematic model, stroke usage and workspace (a 200 mm cube, 8.6 dm ³).
Electrical system	Delivery 1: motor selection (NEMA 34 86HB250, 4.5 N m); Delivery 3, Sheet 3: required torque is 7.1 % of that.

Note on the sheets that follow. The CAD sheets are reproduced as issued in the original submission, so their title blocks are in Spanish and carry the revision data of that submission. The reading

key is: *Materia* = course, *Proyecto* = project, *Dibujó* = drawn by, *Verif.* = checked, *Aprob.* = approved, *Escala* = scale, *Unidades* = units, *Cantidad* = quantity, *Formato* = sheet size, *Material* = material, *Hoja 1 de 1* = sheet 1 of 1, *Las cotas se expresan en mm* = all dimensions in millimetres, *No cambie la escala* = do not scale the drawing, *Romper aristas* = break sharp edges.